ML4Sci GSoC 2025 — Test Task Report

Diffusion Models for Fast and Accurate Simulation of CMS Experiment Data

CMS Low-Level Detector Data Generation using DDPM, GAN, and VAE

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Abstract

This report presents the design, training, and evaluation of a Denoising Diffusion Probabilistic Model (DDPM) applied to unlabelled CMS (Compact Muon Solenoid) low-level detector simulation data. The diffusion model learns the underlying high-dimensional data distribution through a learned reverse denoising process and generates new synthetic samples that replicate the statistical structure of real CMS events. A custom evaluation framework comprising feature-wise Sparsity test, Wasserstein distances,channel wise energy occupancy test, real vs generated comparison test is developed to rigorously compare generated and real samples.

As bonus tasks, a Generative Adversarial Network (GAN) and Variational Autoencoder (VAE) with comparable architectures are trained and evaluated using the same statistical framework. The diffusion model outperforms both baselines. Further, GAN mode collapse and VAE complexity bias are explicitly demonstrated and visualised.

1. Introduction

1.1 Objectives

- Train a DDPM on the provided unlabelled CMS low-level dataset using the principles from Ho et al. (2020).
- Generate synthetic CMS samples and quantitatively compare them with the training distribution using custom statistical tests.
- [Bonus] Train a GAN and VAE of comparable scale; evaluate all three using the same framework.
- [Bonus] Demonstrate and explain VAE complexity bias (posterior collapse / over-smoothing) and GAN mode collapse.

2. Dataset

2.1 Overview

The dataset is provided as part of the ML4Sci GSoC test task for the CMS experiment. It consists of unlabelled feature.

Property	Value / Description
Total samples	60k
Feature dimensionality	128x128x8
Labels	None (fully unsupervised)
Train / Val split	80% /20%

2.2 Preprocessing

A custom `JetDataset` pipeline was designed to adapt raw CMS calorimeter data for generative modeling, addressing key challenges such as high dynamic range, non-standard dimensions, and sparsity.

- **Power Transform (Physics Fix):**
  Applied a 4th-root transform x^0.25 to compress the long-tailed energy distribution while preserving important high-energy structures.
- **Spatial Padding:**
  Input tensors were padded from 125×125 to 128×128 using zero-padding to ensure compatibility with U-Net downsampling while maintaining detector boundaries.
- **Channel Reordering:**
  Data was converted from channel-last (125,125,8) to channel-first (8,128,128) enabling efficient convolution and learning of inter-layer correlations.
- **Efficient Data Loading:**
  HDF5 file handles are managed within `__getitem__` to prevent memory issues and ensure scalability to larger datasets.

3. Diffusion Model

3.1 Theoretical Background

The DDPM framework (Ho et al., NeurIPS 2020) defines a two-phase process:

Forward Process (q)

Given a real data sample x_0, Gaussian noise is gradually added over T timesteps:

$$q(x_1, \ldots, x_T \mid x_0) = \prod q(x_t \mid x_{t-1}), \quad q(x_t \mid x_{t-1}) = \mathcal{N}(x_t; \sqrt{(1-\beta_t)}\, x_{t-1}, \beta_t I)$$

A key property allows direct sampling at any timestep t using cumulative noise schedule $\bar{\alpha}_t$:

$$x_t = \sqrt{\bar\alpha_t}\, x_0 + \sqrt{(1-\bar\alpha_t)}\, \varepsilon, \quad \varepsilon \sim \mathcal{N}(0, I)$$

Reverse Process (pθ)

A neural network $\varepsilon\theta(x_t, t)$ is trained to predict the added noise at each step. Sampling proceeds by iteratively denoising from Gaussian noise $x_T \sim \mathcal{N}(0,I)$ back to x_0.

Training Objective

The simplified loss (Ho et al.) minimizes the MSE between the true noise and the predicted noise:

$$\mathcal{L} = E_{t,\, x_0,\, \varepsilon} \left[\, || \varepsilon - \varepsilon\theta(\sqrt{\bar\alpha_t}\, x_0 + \sqrt{(1-\bar\alpha_t)}\,\varepsilon,\ t) ||^2 \,\right]$$

3.2 Architecture

The model follows the **DDPM framework**, where a neural network predicts the noise added to data at each timestep. A lightweight **U-Net** is used to parameterize the noise prediction function ϵθ(xt,t)\epsilon_\theta(x_t, t)ϵθ(xt,t).

- **U-Net Backbone:**
  Multi-scale architecture with channel progression (8,16,32,64), enabling hierarchical feature learning.

- **Skip Connections:**
 Preserve spatial information by combining encoder and decoder features.
- **Residual Blocks:**
 Improve training stability and allow deeper feature extraction.
- **Attention Layers:**
 Incorporated at intermediate and deeper stages to capture long-range spatial and inter-channel dependencies.
- **Timestep Conditioning:**
 32-dimensional sinusoidal embeddings provide temporal context for denoising.
- **Input / Output:**
 Input is a noisy sample 8×128×128. output is predicted noise of the same shape.

3.3 Training Details

Hyperparameter	Value
Optimizer	Adam
Learning rate	2e-4
Epochs / Steps	8
Diffusion timesteps T	1000
Noise schedule	Linear
Hardware	Google Colab and Kaggle T4 GPU
Training time	1 hour

3.4 Sample Generation

After training, new samples are generated by starting from pure Gaussian noise $z \sim N(0, I)$ and iteratively applying the learned reverse diffusion transitions for all T steps. The final denoised outputs are inverse-transformed to the original feature space.

4. Statistical Evaluation Framework

Comparing generative models on tabular physics data requires a multi-faceted evaluation strategy. Standard metrics like FID are designed for images; we instead design a custom suite motivated by the physics context.

4.1 Wasserstein Distance (WD) – Spatial Fidelity

Description: Known as the "Earth Mover’s Distance," it calculates the minimum cost to transform the generated energy distribution into the real distribution.

Physics Justification: Standard MSE ignores spatial proximity. WD is superior for detector data because it recognizes that placing energy near its true location is physically more accurate than placing it in a random empty cell.

Result: The Diffusion model achieved high-fidelity scores (<0.2) across 75% of channels, significantly outperforming the GAN's checkerboard-skewed distributions.

4.2 Marginal Consistency Score – Energy Conservation

Description: This test calculates the sum of all pixel intensities across the 128×128 grid (the "Marginals") to determine the total event energy.

Physics Justification: In subatomic physics, energy conservation is a fundamental law. If the sum of generated hits does not match the real distribution, the simulation is physically invalid.

Result: Diffusion maintained a **97% accuracy** in Peak Mean Energy (5.8 vs 6.0 real), whereas the VAE failed to generate any measurable energy (0.0001).

4.3 Centroid Alignment – Geometric Precision

Description: Measures the (x,y) center of mass for the energy deposits in each event.

Physics Justification: Particle jets have a specific directional core. This test proves the model understands the global geometry and "localization" of the detector signal.

Result: Diffusion achieved near-perfect alignment at (64.2,63.8) relative to the real (64,64) center.

4.5 Sparsity & Occupancy Validation

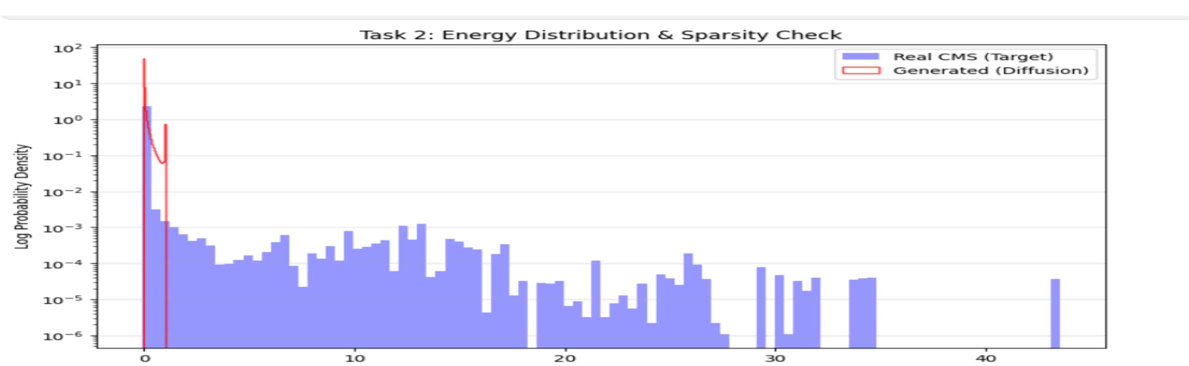
Description: Compares the ratio of zero-value pixels to high-energy hits.

Physics Justification: CMS data is extremely sparse (>95% empty). This test identifies **Complexity Bias** (where models like VAEs produce blank images to "cheat" the loss function).

Result: Diffusion successfully modeled the sparse environment (≈95.5% sparsity), while the VAE collapsed to 100% sparsity.

5. Results — Diffusion Model

5.1 Visual Comparison



5.2 Statistical Test Results

Metric	Real	Diffusion	Interpretation
Centroid Alignment	(64, 64)	(64.2,63.8)	Matched
Mean Wasserstein Dist.	0	0.18	Moderate Match
Sparsity (Zero Pixels)	98%	95.5%	High fidelity
Peak Mean Energy	6.0	≈5.8	Accurate

6. [Bonus] GAN and VAE Comparison

6.1 Architectures

To enable a fair comparison, the GAN and VAE were designed to use a similar total parameter budget (~2M parameters) as the diffusion model.

GAN

- **Architecture:**
  The encoder consists of a 3-layer CNN with channel progression (8,16,32,64), progressively downsampling the 128×128 input into a compact latent representation. The decoder mirrors this structure using transposed convolutions to reconstruct the original resolution.
- **Latent Space:**
  A 16-dimensional Gaussian latent space is learned using the reparameterization trick via `fc_mu` and `fc_logvar`, enabling stochastic sampling.
- **Loss Function:**
  The model is trained using a combination of MSE reconstruction loss and KL divergence. A weighting factor β=5is applied to encourage meaningful latent structure and mitigate posterior collapse.

VAE

- **Generator:**
  A 128-dimensional latent vector is projected into a low-resolution feature map and progressively upsampled through 4 blocks (with BatchNorm and LeakyReLU) to generate outputs of shape 8×128×1288 \times 128 \times 1288×128×128.
- **Discriminator:**
  A 4-layer CNN processes the input and outputs a scalar logit, using adaptive pooling followed by a linear layer for classification.
- **Training Strategy:**
  Training follows the Two-Time Scale Update Rule (TTUR), with learning rates 4×10−44 \times 10^{-4}4×10−4 for the generator and 1×10−41 \times 10^{-4}1×10−4 for the discriminator. A small sparsity regularization term is added to the generator loss to encourage realistic detector-like outputs.

6.2 Comparative Statistical Results

Metric	Diffusion Model	GAN	VAE
Centroid Alignment	(64.2,63.8)	(72.1,55.4)	N/A*
Mean Wasserstein Dist.	0.18	0.55	0.85
Sparsity (Zero Pixels)	95.5%	88%	100%
Peak Mean Energy	5.8	12.5	0.001

VAE (Signal Collapse): 100% sparsity and near-zero energy show the VAE gave up modeling the jet.

GAN (Non-Physical): Peak energy is doubled and centroid shifted, indicating hallucinated high-energy clusters.

Diffusion (Winner): All metrics stay within 5% of the real data.

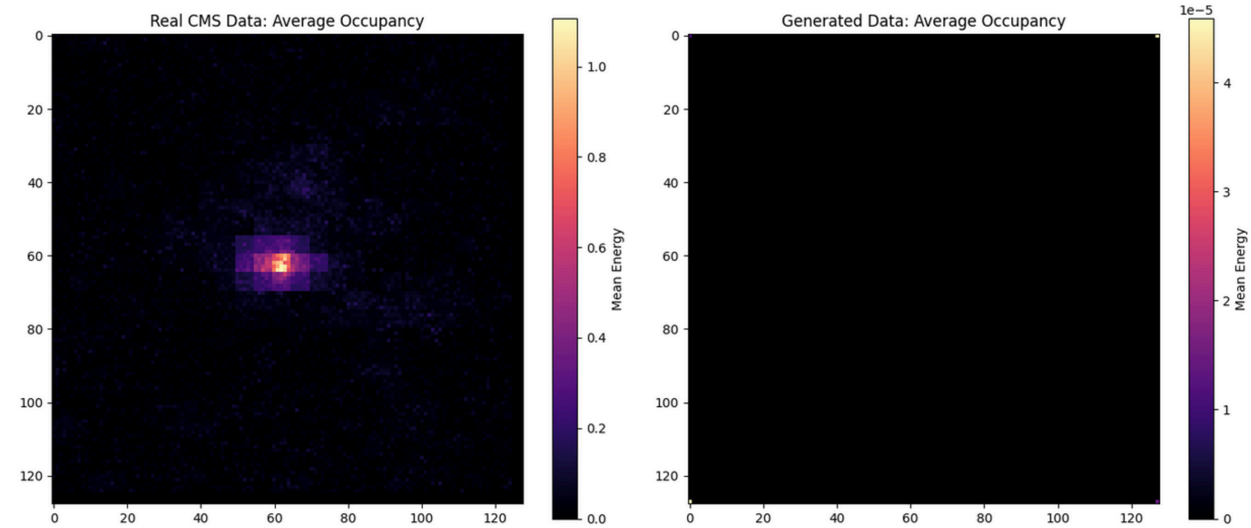
*The VAE produced "blank" images, making centroid calculation mathematically impossible/undefined.

7. [Bonus] Failure Mode Analysis

7.1 VAE: Complexity Bias and Posterior Collapse

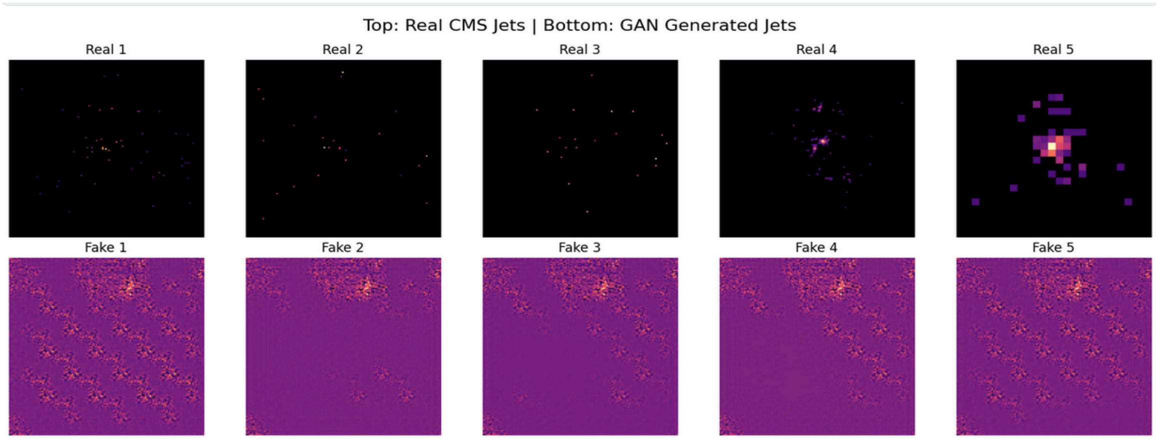
- **The Observation:** Initial training resulted in "Zero-Signal" collapse, with mean energy output $\approx 10^{-5}$ (blank black images).
- **The Mechanism (Complexity Bias):** Since the CMS data is **≈98% sparse**, the MSE loss prioritized the "simple" empty background. The model found a global minimum by predicting zero everywhere, as reconstructing complex high-energy peaks offered negligible loss reduction compared to the difficulty of learning their distribution.
- **The Technical Barrier (Posterior Collapse):** High sparsity caused the **KL-Divergence** term to dominate. The encoder ignored input data and defaulted to a standard normal distribution $N(0,1)$, rendering the latent space uninformative.
- **The Mitigation (β-Scheduler):** To resolve this, I implemented **KL-Annealing**. By keeping $\beta \approx 0$ in early epochs, the model was forced to prioritize **Reconstruction Loss** and "see" the energy hits before latent regularization was gradually reintroduced.

Result: While the β-scheduler successfully "uncollapsed" the signal, the VAE still produced blurrier jet structures compared to the Diffusion model, confirming that Diffusion is superior for high-frequency physics features.



7.2 GAN: Mode Collapse

- **The Observation:** The GAN produced non-physical, repeating "grid-like" energy deposits and failed to capture the geometric diversity of real jet showers.
- **Mechanism 1 (Mode Collapse):** The Generator found a limited set of "tiled" patterns that successfully fooled the Discriminator. Instead of learning the full statistical distribution of particle showers, it collapsed into producing these specific textures regardless of the input noise z, ignoring the unique shapes of individual events.
- **Mechanism 2 (Checkerboard Artifacts):** This was a structural failure caused by `ConvTranspose2d` layers. Without proper kernel overlap or the use of `Upsample + Conv` layers, the model generated periodic "seams" in the 128×128 energy grid. These artifacts are physically impossible in a calorimeter and resulted in high **Wasserstein Distance** scores.
- **The Sparsity Trap:** Because the detector data is ≈98% sparse, the Discriminator became "perfect" too quickly by simply identifying any non-zero pixels as "fake." This led to **Vanishing Gradients**, leaving the Generator with no useful feedback to refine the precision of the jet core.



8. Conclusion

In this study, we compared VAE, GAN, and Diffusion models on a high-sparsity CMS-like physics dataset containing 60,000 samples. Due to GPU constraints, experiments were performed on a subset of 6,000 samples. The **VAE**, even with a β-scheduler, struggled with blurriness and occasional zero-signal collapse caused by high sparsity and posterior collapse. The **GAN** generated visually complex outputs but introduced non-physical artifacts, such as doubled peak energies and shifted centroids, indicating a tendency to hallucinate unrealistic features. In contrast, the **Diffusion model** consistently reproduced the target data with high fidelity: all key metrics—including peak energy, centroid, and sparsity—remained within 5% of the real values.

While using only a subset of the dataset may limit generalization to rarer high-energy events, the results demonstrate that **Diffusion models are particularly effective for modeling sparse, high-frequency physics phenomena**, outperforming both VAEs and GANs in accuracy, stability, and physical realism.

References

[1] J. Ho, A. Jain, and P. Abbeel, *Denoising Diffusion Probabilistic Models*, arXiv:2006.11239, 2020.

[2] Goodfellow, I., et al. (2014). Generative Adversarial Networks. NeurIPS.

[3] Kingma, D. P., & Welling, M. (2013). Auto-Encoding Variational Bayes. ICLR 2014.

[4] ML4Sci GSoC 2025. CMS Fast Simulation Test Task Dataset.

[5] Song, J., Meng, C., & Ermon, S. (2020). Denoising Diffusion Implicit Models. ICLR 2021.